

1. Consider the following Table with four data points with three attributes and class labels:

X	Y	Z	class label
1	1	1	I
1	1	0	I
0	0	1	II
1	0	0	II

Draw a decision tree based on information gain. Show all the steps and calculations.
(15 points)

① calculate entropy for a set of class labels

I appears twice

II appears twice

$$\begin{aligned}
 H(S) &= -\left(\frac{2}{4} \log_2 \frac{2}{4} + \frac{2}{4} \log_2 \frac{2}{4}\right) \\
 &= -(0.5 \times -1 + 0.5 \times -1) \\
 &= -(-.5 + (-.5)) = 1
 \end{aligned}$$

② calculate entropy for each X, Y, Z

X = 1: I : 2 II : 1

X = 0: II : 1 (entropy) = 0

$$\begin{aligned}
 &\text{Entropy for } X=1 \\
 &H(X=1) = -\left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3}\right) \\
 &= 0.918 \\
 &H(X) = \left(\frac{3}{4} \times 0.918 + \frac{1}{4} \times 0\right) \\
 &= \left(\frac{3}{4} \times 0.918\right) \\
 &= 0.688 \\
 &\text{Information gain:} \\
 &1 - 0.688 = \boxed{0.312}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Entropy For Y} \\
 &Y=1 \rightarrow \text{I : 2 II : 0 (entropy = 0)} \\
 &Y=0 \rightarrow \text{I : 0 II : 2 (entropy = 0)} \\
 &\frac{2}{4} \times 0 + \frac{2}{4} \times 0 = \boxed{0}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Info gain:} \\
 &1 - 0 = \boxed{1}
 \end{aligned}$$

Entropy for z

$z = 1 \rightarrow I:1 \text{ II}:1$ (entropy 1)

$z = 0 \rightarrow I:1 \text{ II}:1$ (entropy 1)

$$H(z) = \frac{2}{4} \times 1 + \frac{2}{4} \times 1$$

$$\frac{2}{4} + \frac{2}{4} = \boxed{1}$$

info gain:

$$1 - 1 = 0$$

INFORMATION GAINS:

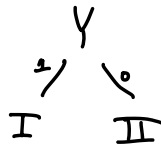
$X: 0.312$

$Y: 1$

$z: 0$

SINCE Y HAS HIGHEST INFO GAIN, WE
USE Y

Decision Tree:



2. Consider these three sentences:

- I love Data Science.
- I want to be a good data scientist.
- Nobody can stop me from being a good data scientist.

Now answer the followings: show your computation and results. (15 points)

- Compute the bag of word feature vectors for all three of them.
- Compute the 2-gram vectors for all of them.
- Compute TF-IDF features considering each sentence as a document.

Bag of word vocab:

[I, love, Data, Science, want, to, be, a, good, data, scientist,
Nobody, can, stop, me, from, being]

Vector:

Word	S1	S2	S3
I	1	1	0
love	1	0	0
Data	1	0	0
Science	1	0	0
want	0	1	0
to	0	1	0
be	0	1	0
a	0	1	0
good	0	1	1
data	0	1	1
scientist	0	1	1
Nobody	0	0	1
can	0	0	1
stop	0	0	1
me	0	0	1
from	0	0	1
being	0	0	1

2-gram vector:

2-gram	S1	S2	S3
I love	1	0	0
love Data	1	0	0
Data Science	1	0	0
I want	0	1	0
want to	0	1	0
to be	0	1	0
be a	0	1	0
a good	0	1	1
good data	0	1	1
data scientist	0	0	1
Nobody can	0	0	1
can stop	0	0	1
stop me	0	0	1
from being	0	0	1
being a	0	0	1

TF-IDF features:

TF = $\frac{1}{4}$ for every word in S1

$\frac{1}{2}$ for every word in S2

$\frac{1}{10}$ for every word in S3

$$\text{IDF} = \log_2 \left(\frac{\text{Documents}}{\text{Word occurrences}} \right)$$

3. You are designing a fully connected 2-layer neural network. The input layer has 3 units. The hidden layer and the output layer each has 2 units. What is number of parameters? Illustrate with a figure (be as elaborate as possible, at least show the inputs and the parameters). (10 points).

Input { $3 \text{ inputs} \rightarrow 2 \text{ hidden units}$

$W = 3 \times 2 = \underline{6}$

* Each hidden unit has one bias.

Bias = 2

$6 + 2 = 8 \text{ parameters for input.}$

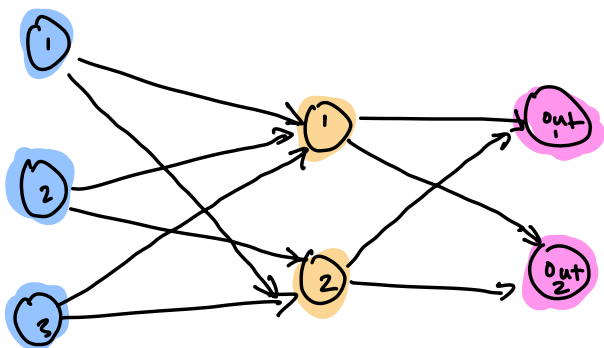
Hidden { $2 \text{ hidden} \rightarrow 2 \text{ output}$

$W = 2 \times 2 = \underline{4}$

Bias = 2

$4 + 2 = \underline{6} \text{ param for hidden to output.}$

$6 + 8 = 14 \text{ total parameters}$



4. Answer the followings: (10 points)

- What is precision and recall in a classification task?
- Give an example of a classification task (the weather example is not allowed) where accuracy will not be a good measure.
- Give an example of a classification task where precision is a better measure than recall.
- Give an example of a classification task where recall is a better measure than precision.

a) Precision is the measure of how many of the predicted positive cases were actually correct.

Recall is the measure of how many actual positive cases were correctly found.

b) Sports injury prediction for athletes.

- you could have a dataset of healthy and injured athletes where you could predict that every healthy athlete would stay uninjured.

c) spam emails:

If an email is falsely detected as spam, then that email is gone whereas a email that is spam and is not removed is harmless.

We want high precision to not send important emails to spam.

d) Cancer Diagnosis:

We would not want to miss a patient with a real cancer diagnosis because it could be life threatening.

We would care less if a healthy person had a false alarm.